IETF-125 I2ICF Side Meeting

# Use Case of I2ICF: AI Agent-Based LivingCare Framework for Human Welfare

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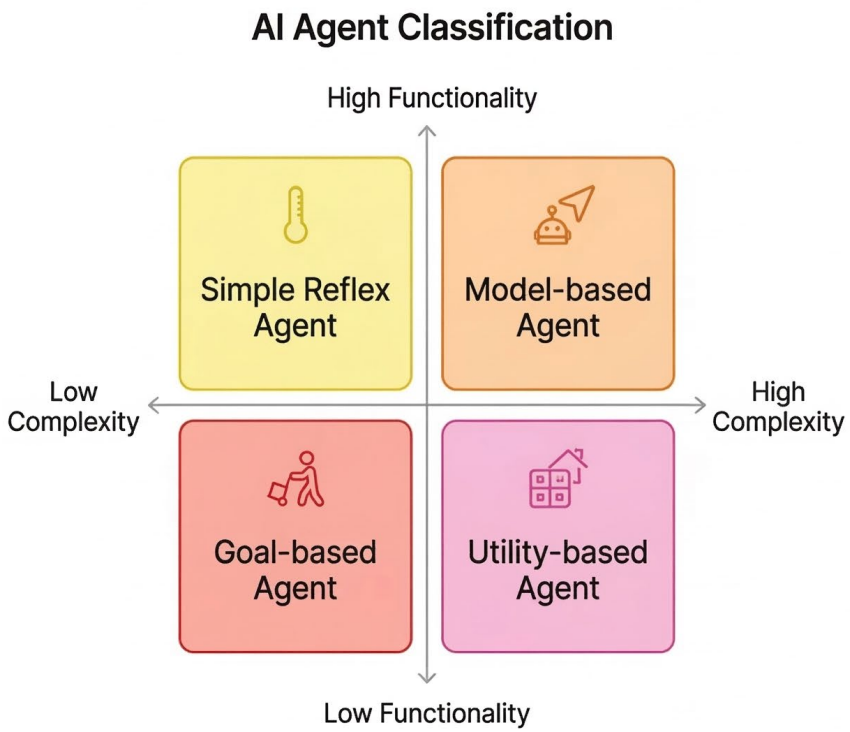
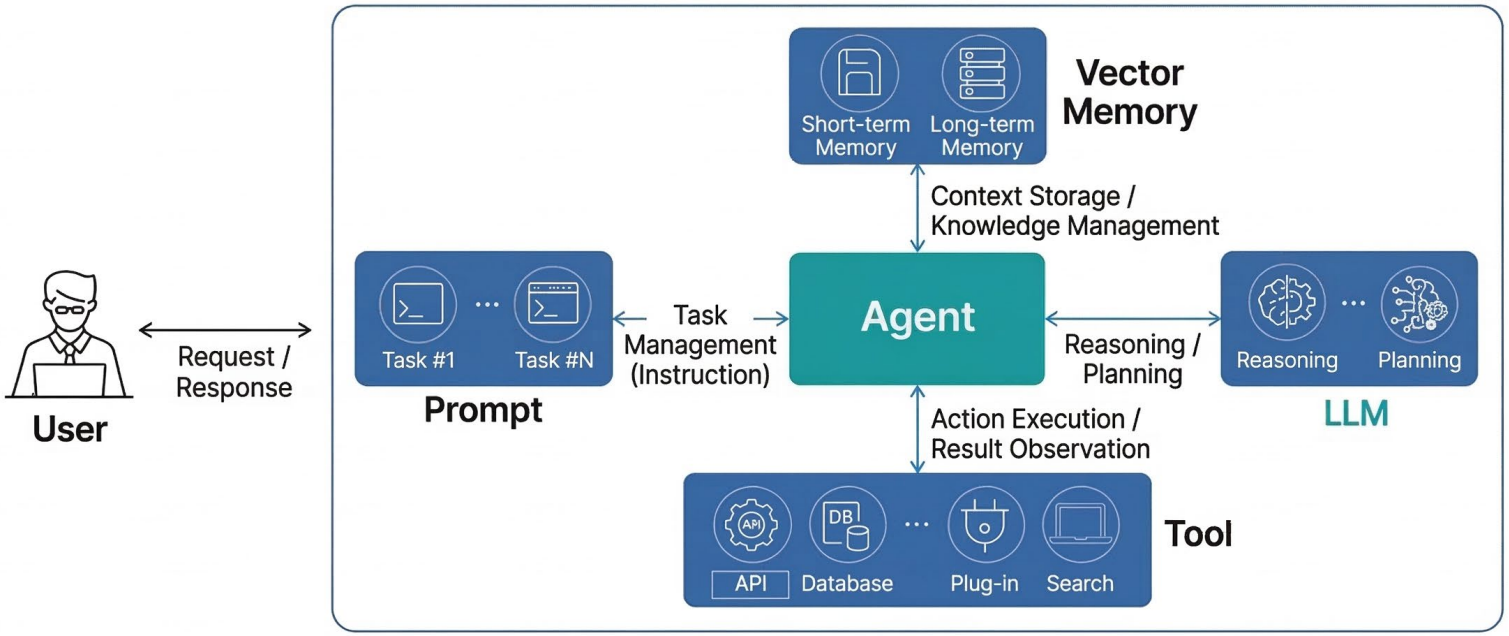
# Background (1/4)

## AI Agent Types and How They Work

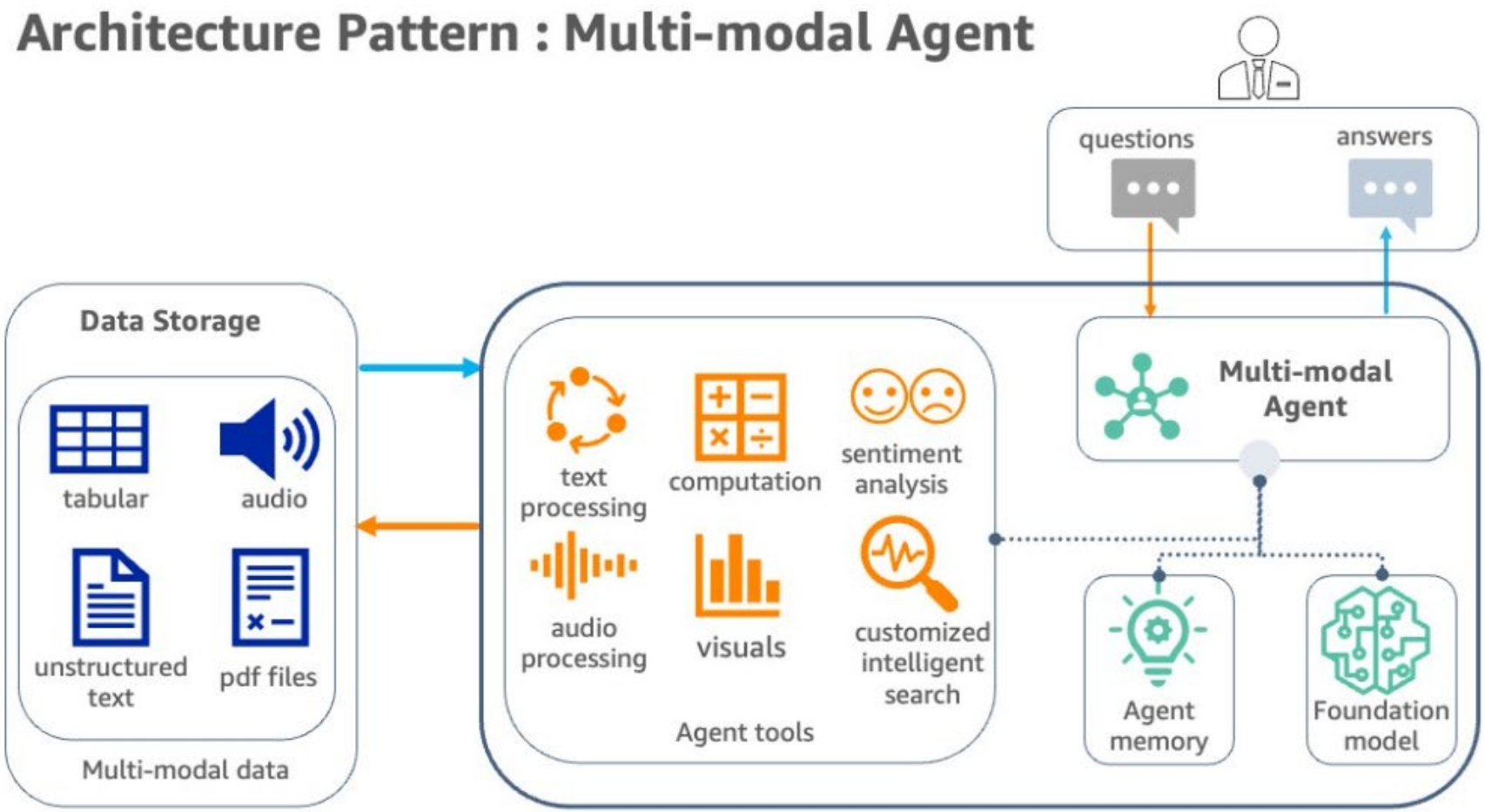



[Source] https://www.samsungsds.com/kr/insights/ai-strategy-for-your-company.html
[Source] https://www.samsungsds.com/kr/insights/integrating-ai-assistants-into-business-workflows-part1.html

# Background (2/4)

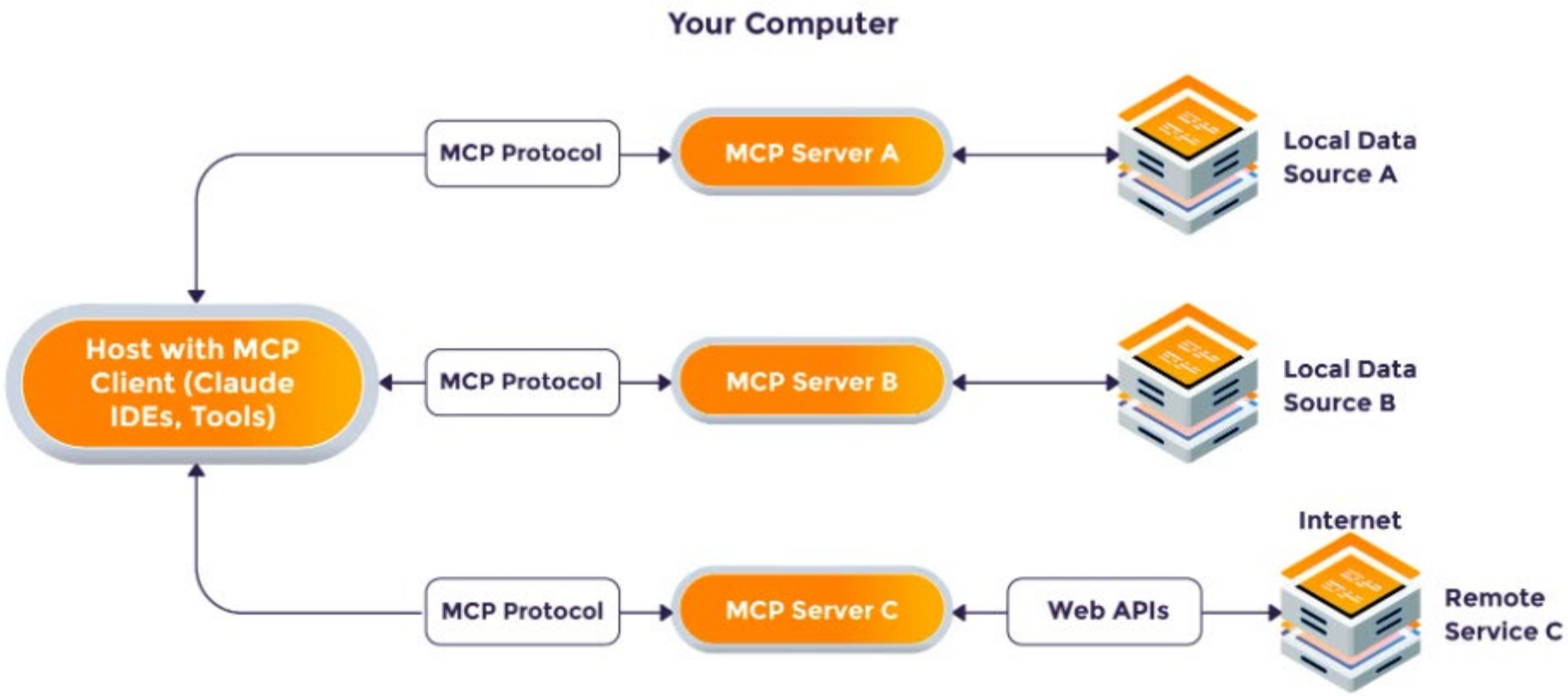
## Architecture Pattern : Multi-modal Agent




[Source] https://www.samsungsds.com/kr/insights/integrating-ai-assistants-into-business-workflows-part1.html

# Background (3/4)

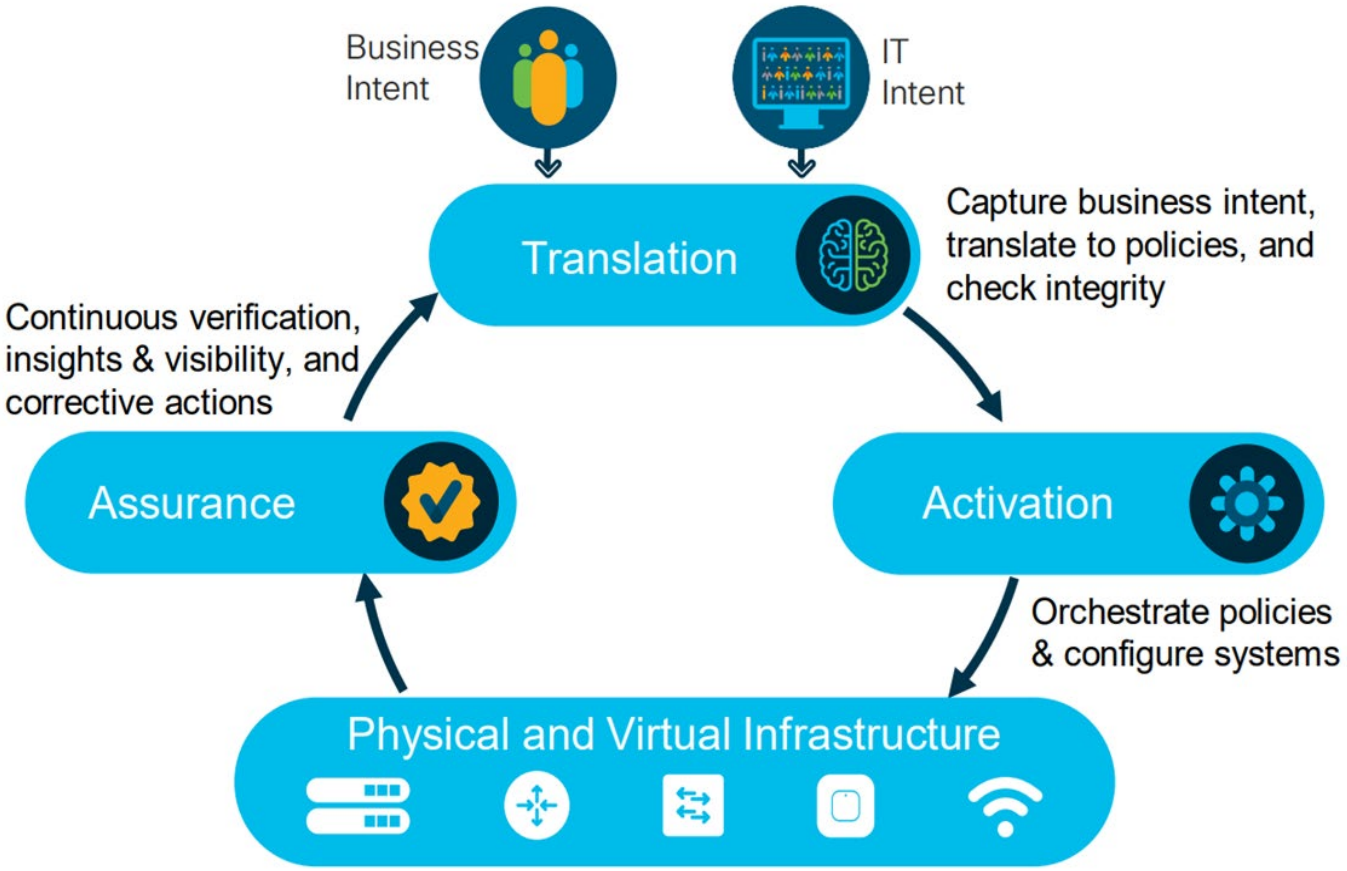
## Model Context Protocol (MCP)




[Source] https://www.searchunify.com/resource-center/short-articles/the-technical-backbone-of-mcp-architecture-how-it-works

# Background (4/4)

## Intent-Based Networking (IBN): Cisco




### Simplified Operations

Centralize network policy control, automate management, and integrate IT workflows

### Proactive Management

Use network monitoring and analytics, aided by machine learning, to pinpoint, report, and remedy issues before they become problems.

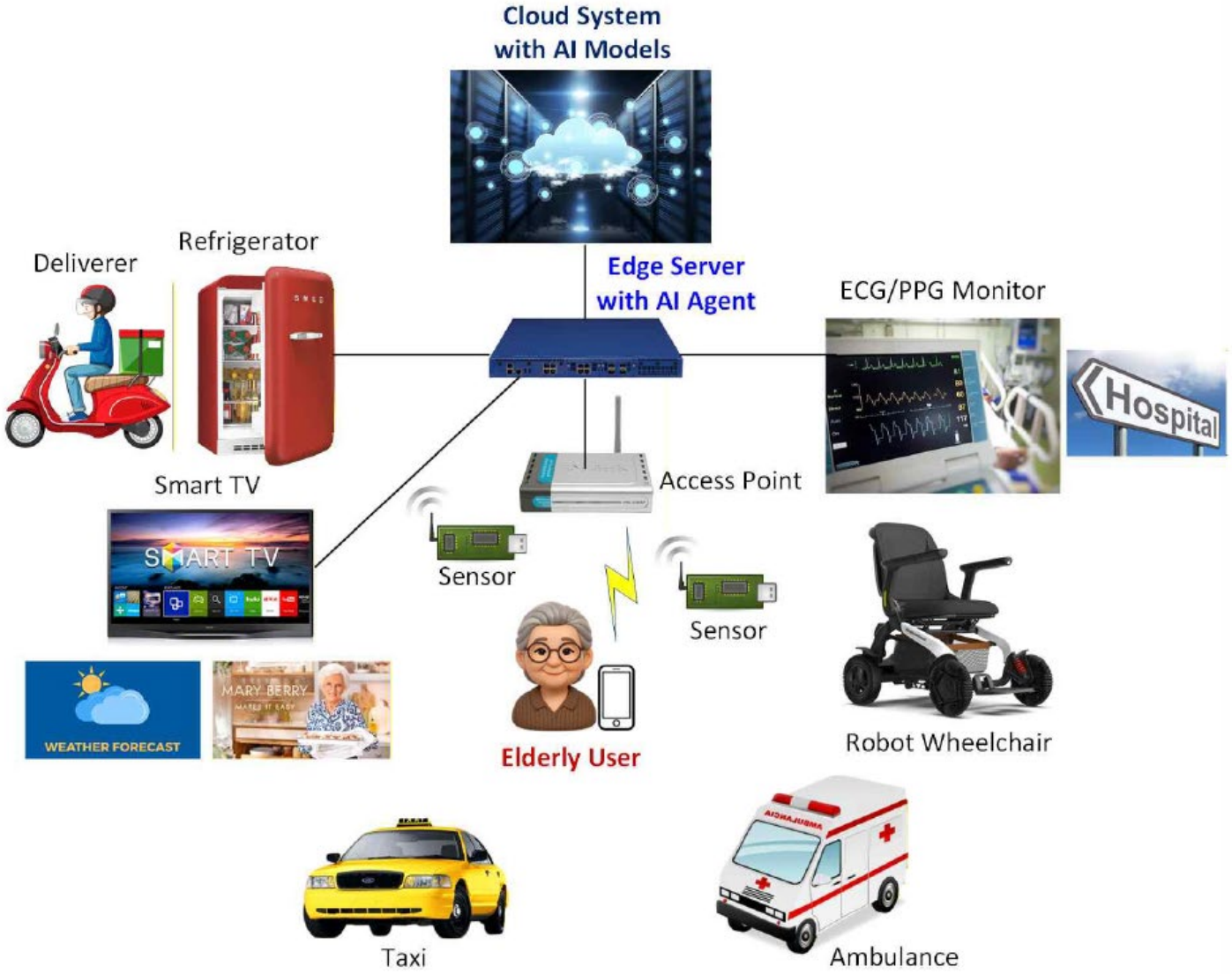
### Security Throughout the network

Detect and contain threats everywhere, in real time. Use network intelligence to quickly alert security operations.

[Source] https://www.cisco.com/c/en_au/solutions/intent-based-networking.html

# AI LivingCare Integrated Network

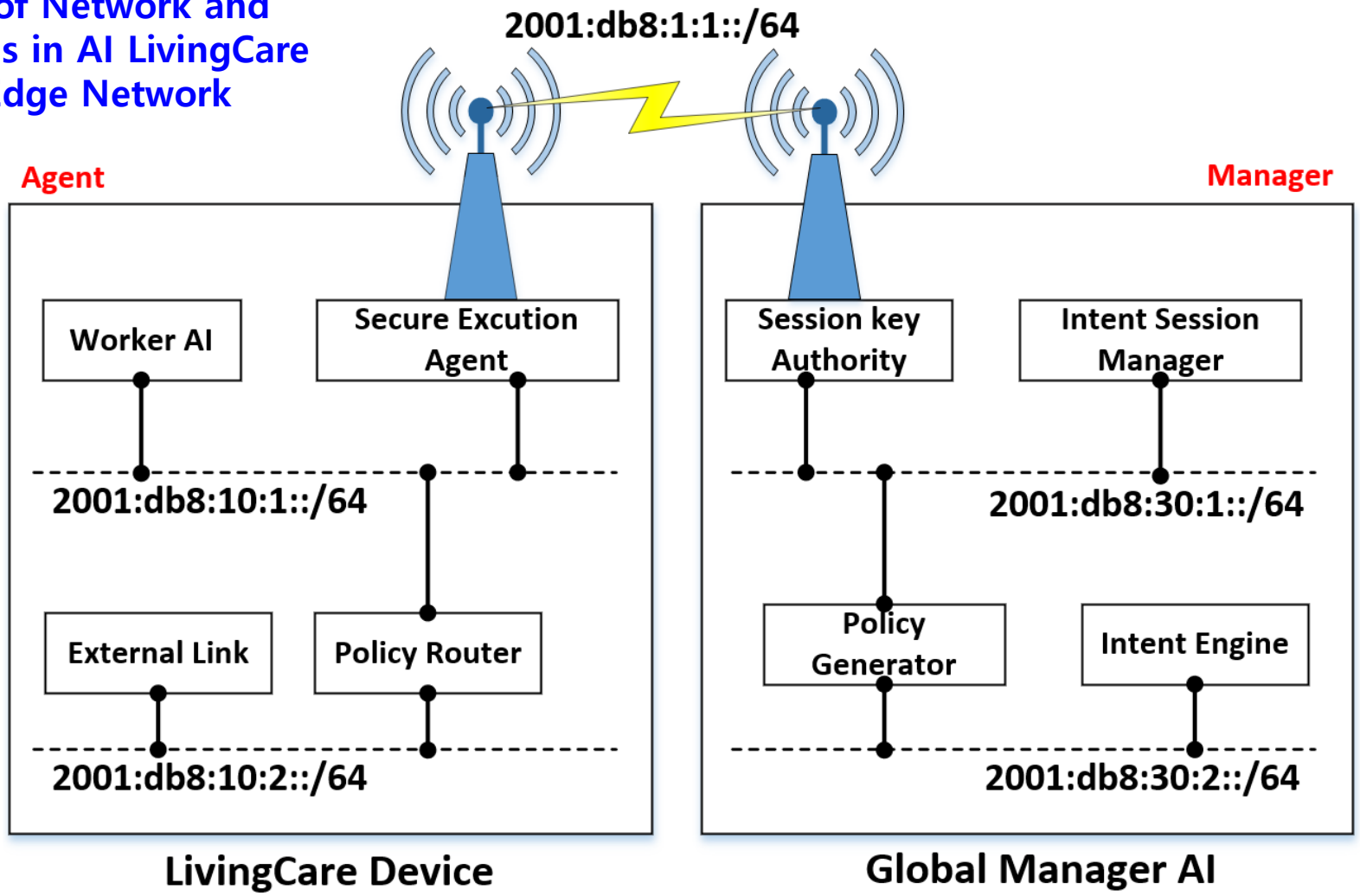
## AI LivingCare Multi-Layer Network Architecture



# AI LivingCare Security Session Control Architecture

## AI LivingCare Intent-Driven Secure Session Control Architecture (Manager-Agent)

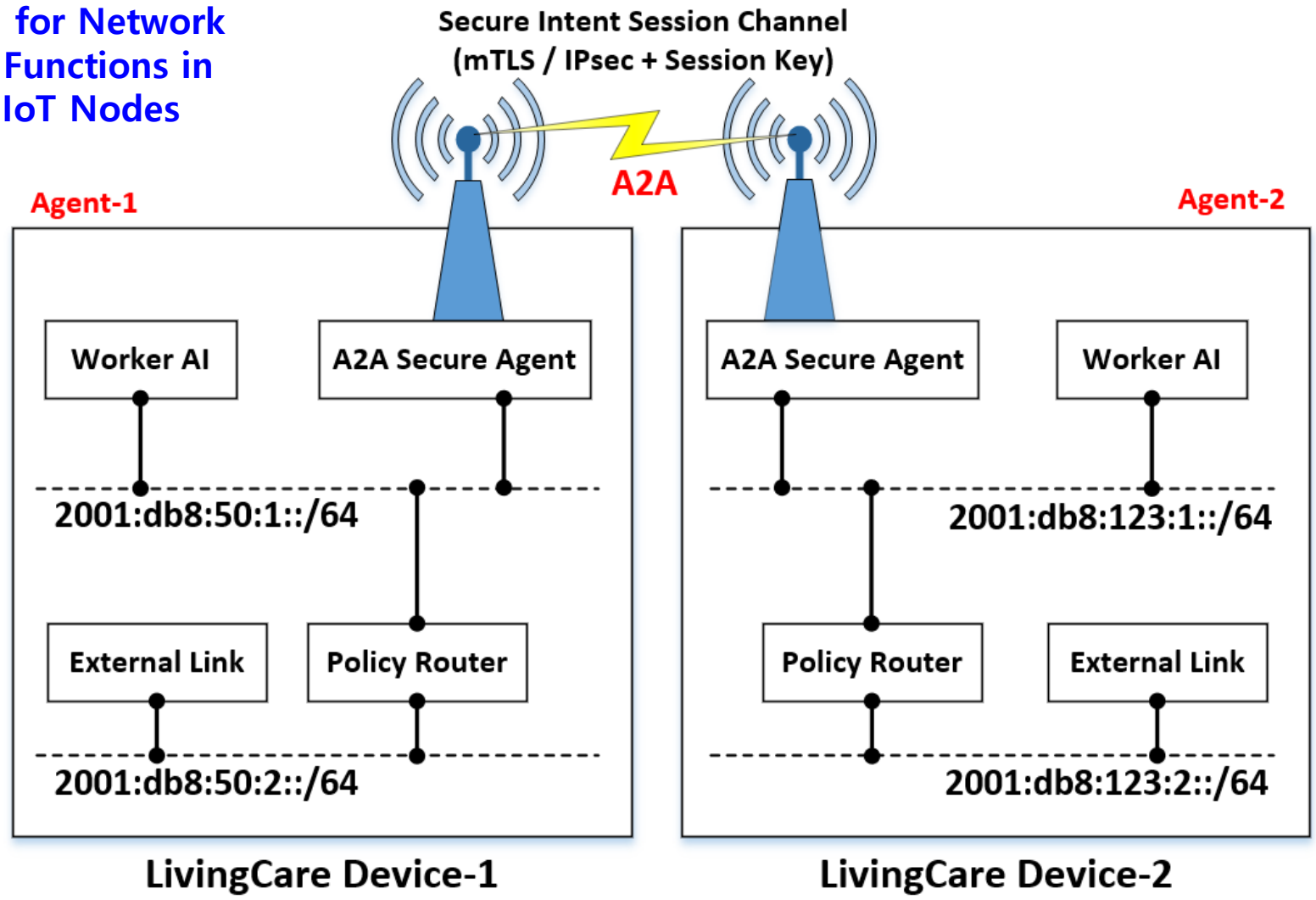
**Configuration of Network and Security Functions in AI LivingCare Device and Edge Network**



# AI LivingCare AI Device Collaboration Architecture

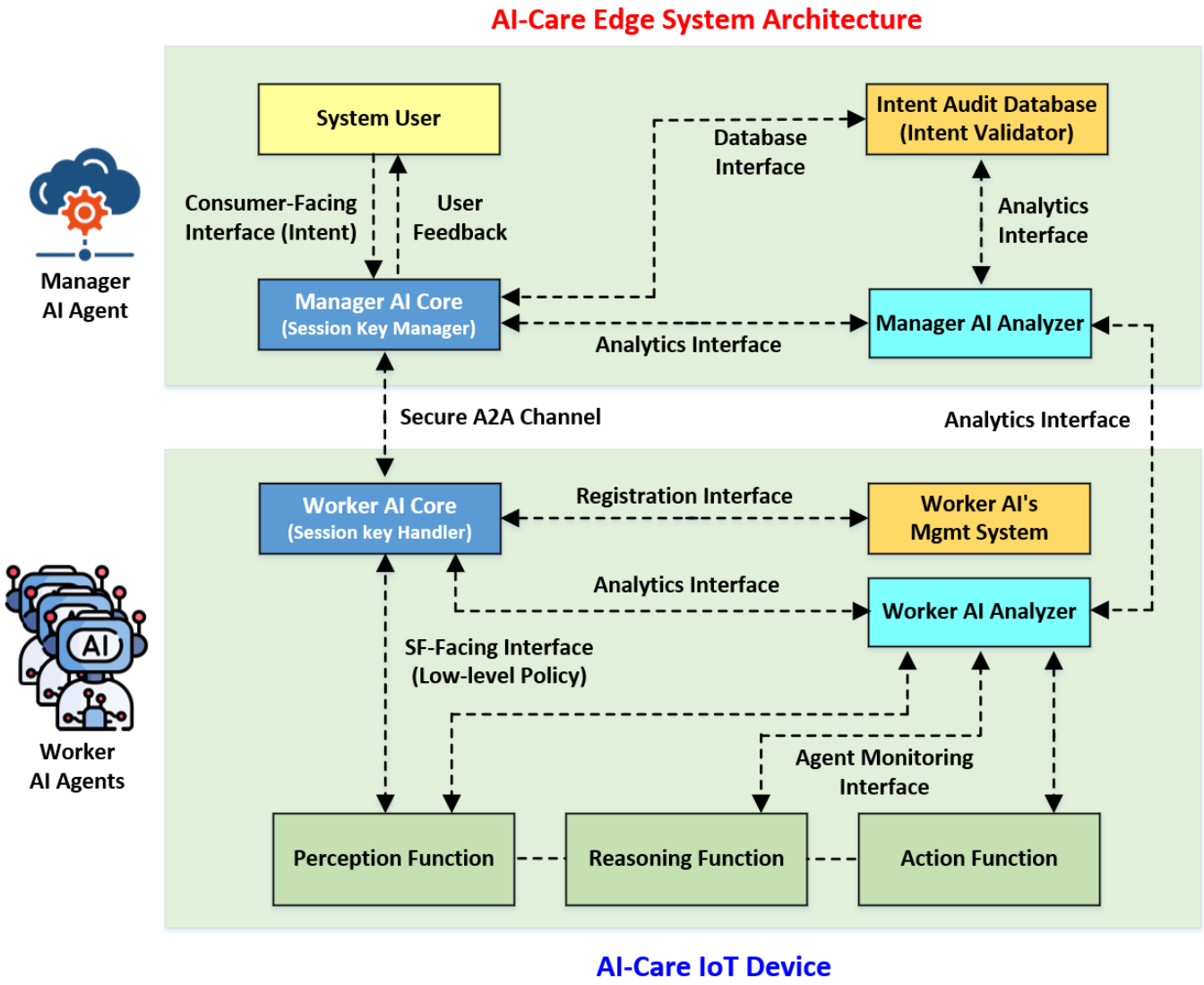
## AI LivingCare IoT Nodes with Distributed Agent Collaboration (Agent-Agent)

**Configuration for Network and Security Functions in LivingCare IoT Nodes**



# AI LivingCare Edge System Framework (1/6)

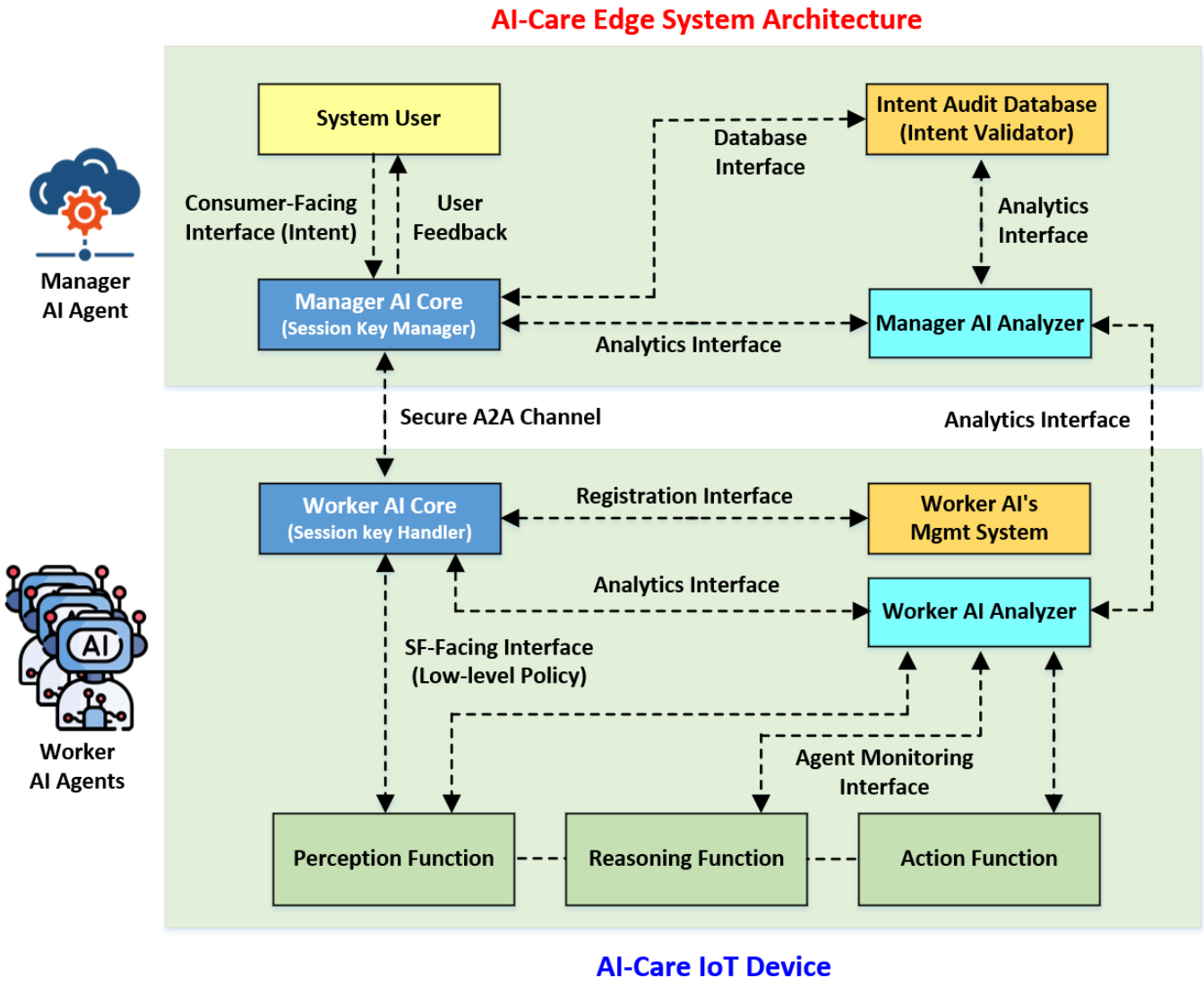
## AI LivingCare Framework with Manager AI and Worker AI




- **Unified Configuration for Networks, Security, and Applications**
  - IPv6 Network Prefixes
  - Default Gateways
  - DNS Servers
  - Network Security Policies
  - Application Parameters
- **Unified Monitoring**
  - Edge AI Agent States
  - Network Functions
  - Service Functions

# AI LivingCare Edge System Framework (2/6)

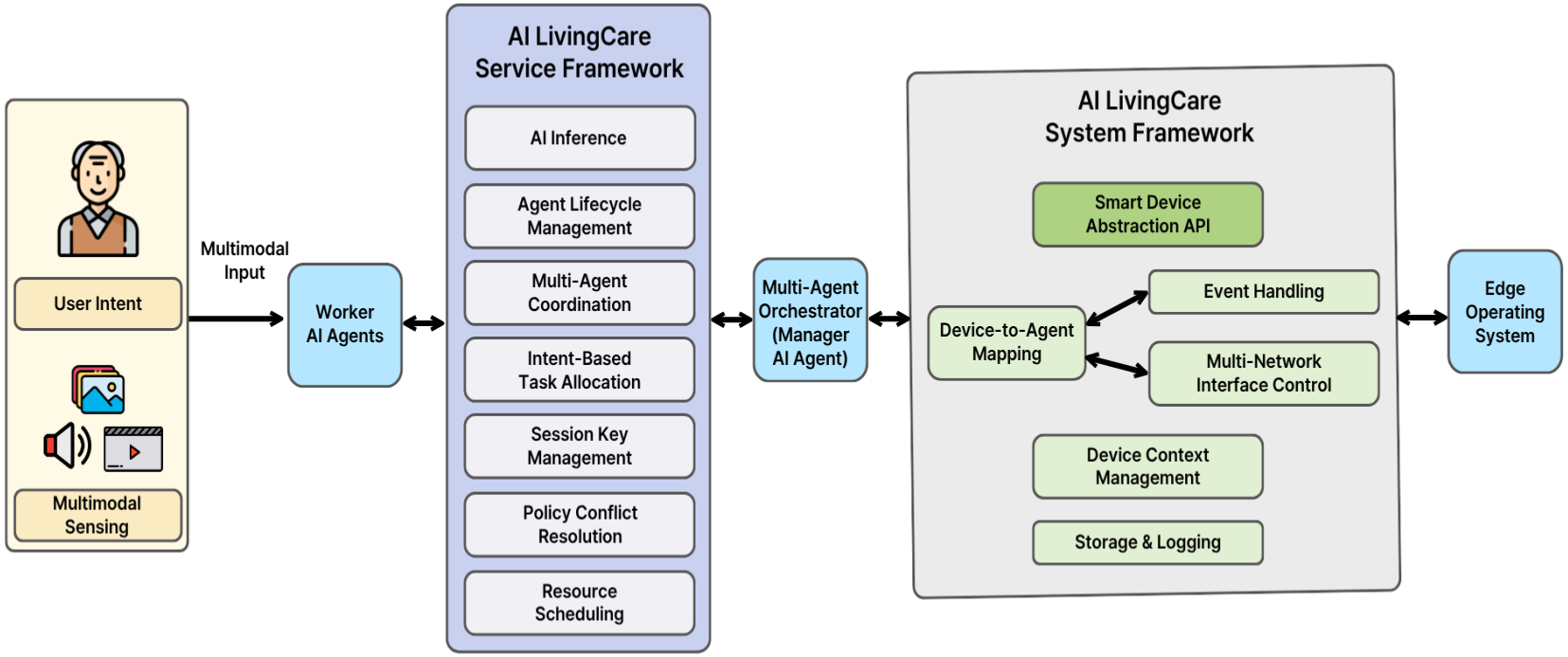
## AI LivingCare Framework with Manager AI and Worker AI




- **Components**
  - Manager AI Agent (Intent & Session Control)
  - Worker AI Agent (Perception-Reasoning-Action)
  - Intent Audit & Policy Management
  - A2A Security Module
- **Interfaces**
  - Communication Messages for two Components
  - Secure A2A Channel: mTLS/IPsec + Session Key
  - Protocols: REST API, NETCONF, and RESTCONF
  - Documents: XML, JSON, and YAML

# AI LivingCare Edge System Framework (3/6)

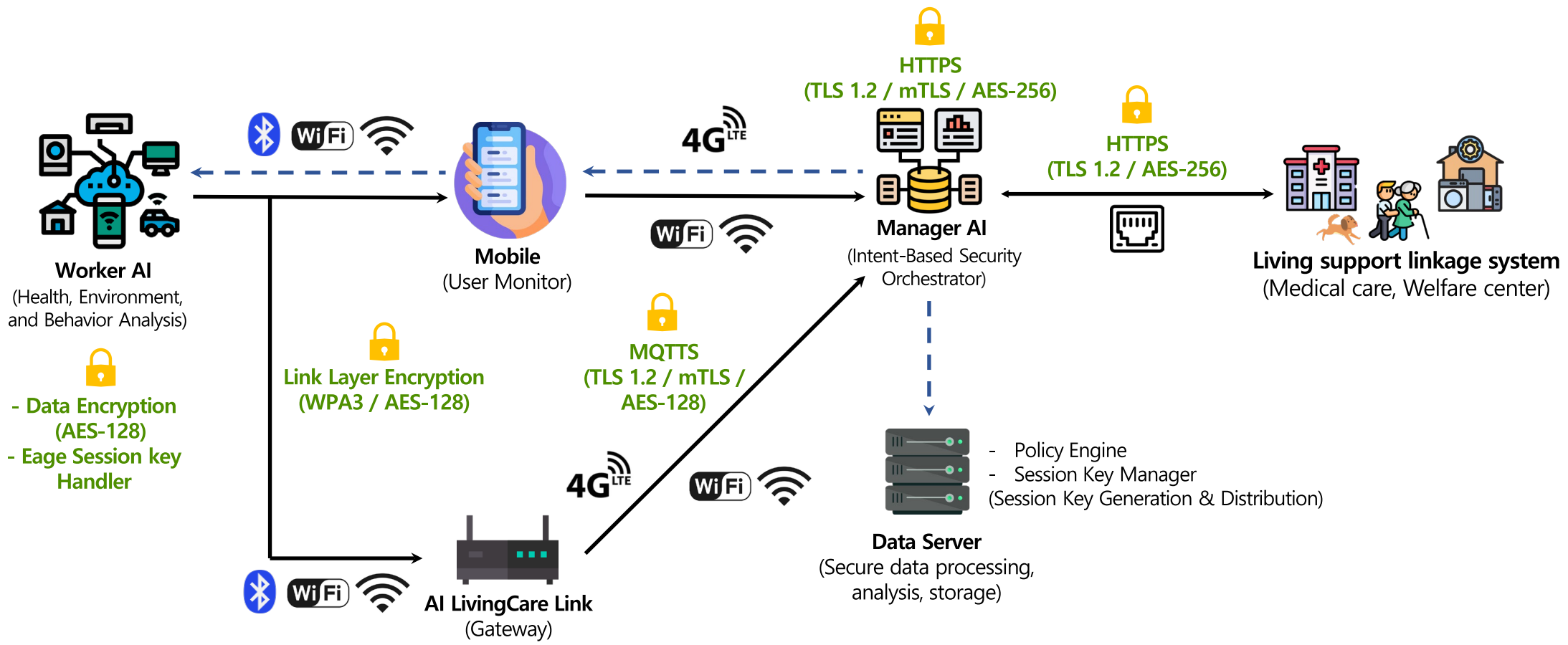
## AI LivingCare Framework Requirements




- Analysis of AI LivingCare Framework Requirements and Policy Translation Functions
  - Analysis of Requirements for Multimodal Input-Based AI LivingCare Platform Framework
  - Analysis of Intent-Based Policy Translation Function Requirements

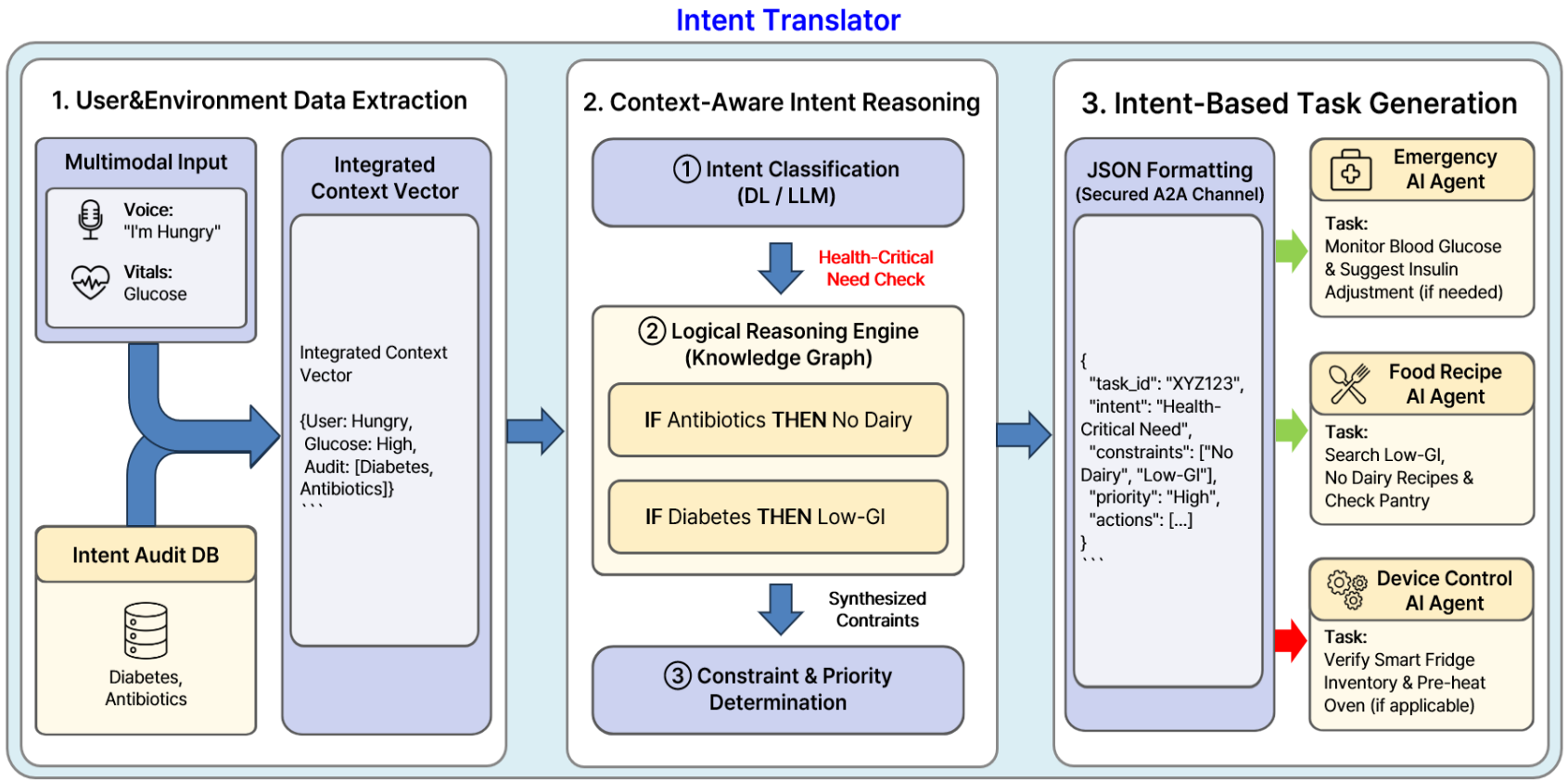
# AI LivingCare Edge System Framework (4/6)

## AI LivingCare Framework Information Security System



# AI LivingCare Edge System Framework (5/6)

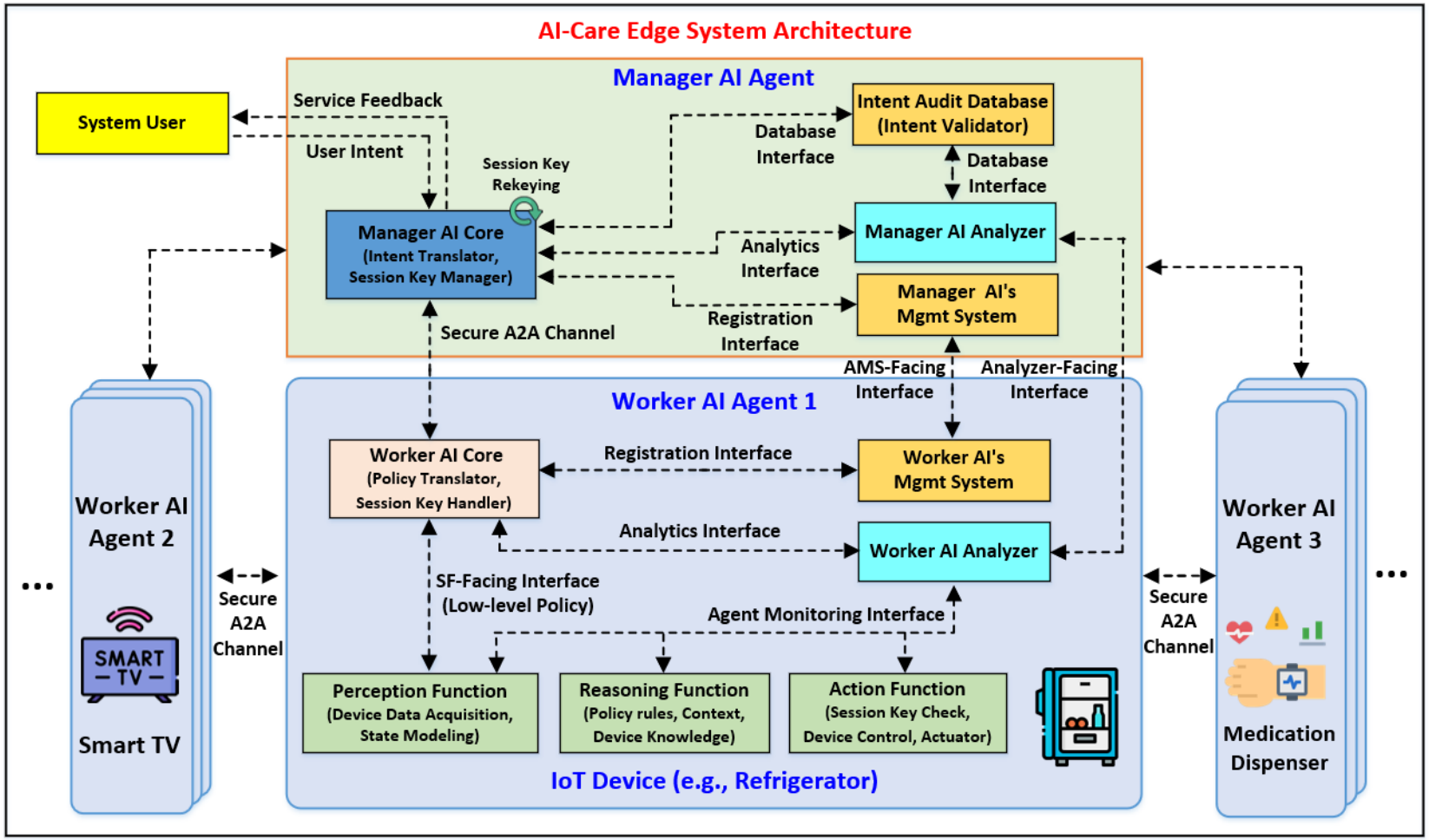
## Intent Translator for AI LivingCare Framework




- Implementation of Intent Translator for AI LivingCare Framework
  - Implementation of Multimodal Input-Based AI LivingCare Framework
  - Implementation of Intent Translator

# AI LivingCare Edge System Framework (6/6)

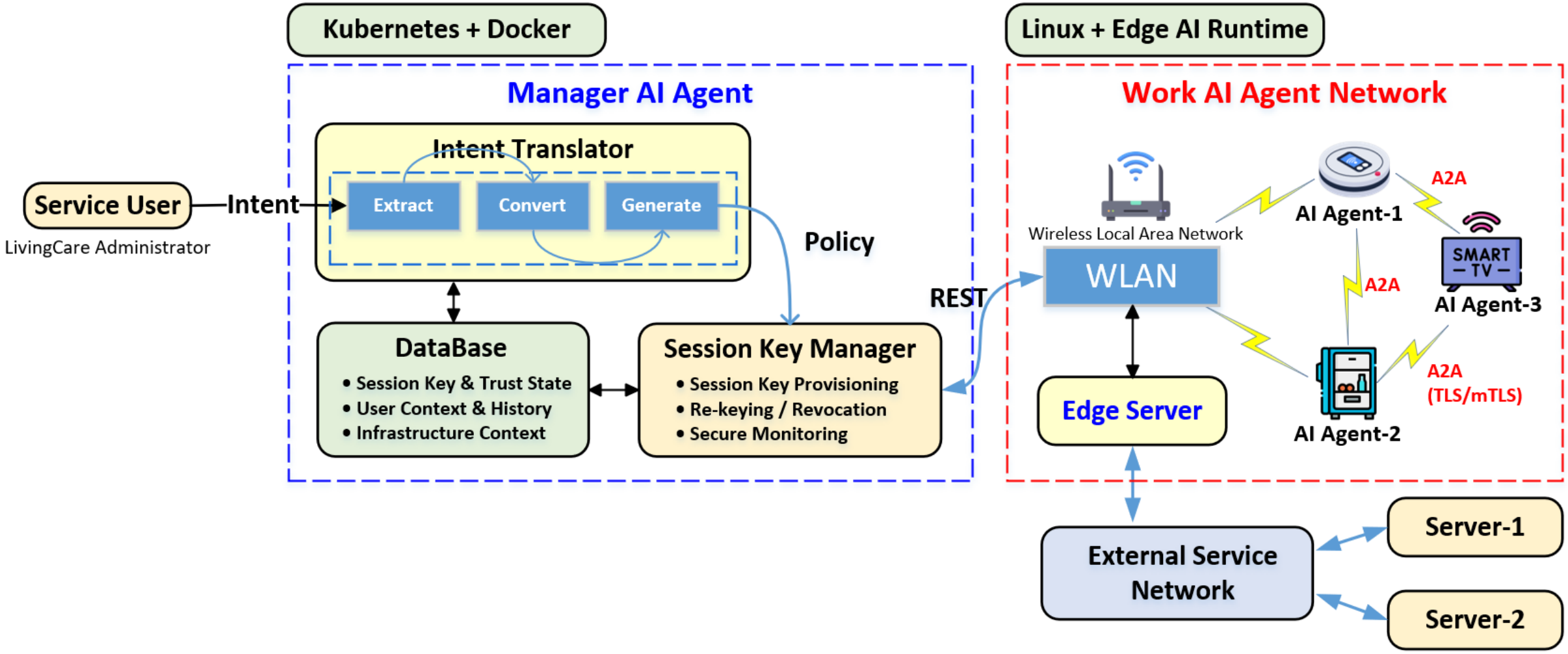
## Multi-AI Agent Collaborative Communication Framework




- AI Agent Function Configuration and Implementation of Multi-Agent Collaborative Communication Framework
  - Implementation of Smart Device Registration and Management System
  - Implementation of Cooperative Communication to ensure Security between Smart Devices

# AI LivingCare Edge System Testbed

## AI LivingCare Framework Testbed



# Next Step

- This work introduces a Use Case of I2ICF for AI-Based LivingCare Framework for Human Welfare with an Intent Translator as follows:
  - An Intent Translation for IoT Devices including Moving Objects:

  https://datatracker.ietf.org/doc/draft-gu-nmrg-intent-translator/

- I2ICF Group will prepare a WG-Forming BoF in the IETF126 in Vienna in July in2026.

- I2ICF Group will prepare IETF-126 Hackathon Project to clarify the main drafts:
  - I2ICF Problem Statement and Use Cases
  - I2ICF Framework

# Thank You and
# Any Questions/Comments?

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